# Dong Heon Han

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Robotics Department, University of Michigan, Ann Arbor, MI, USA

## SUMMARY

A PhD student developing large-scale electromagnetic platforms for microrobot swarms, focusing on collective control, swarmalator dynamics, and programmable morphing

## EDUCATION

- **University of Michigan** — *2029*
  *PhD in Robotics* — Ann Arbor, MI
- **University of Michigan** — *2025*
  *MS in Mechanical Engineering* — Ann Arbor, MI
- **Georgia Institute of Technology** — *2021*
  *BS in Mechanical Engineering* — Atlanta, GA

## WORK EXPERIENCE

- **University of Michigan | Robotics Department** — *June 2025 - Present*
  *Research Assistant | PI: Dr. Steven Ceron* — Ann Arbor, MI
- **University of Michigan | Mechanical Engineering Department** — *Aug 2023 - May 2025*
  *Research Assistant | PI: Dr. Daniel Bruder* — Ann Arbor, MI
- **Republic of Korea Army** — *Jan 2022 - July 2023*
  *Signal Specialist* — Korea
- **Seoul National University | Biosystems Engineering Department** — *Aug 2021 - Dec 2021*
  *Research Assistant | PI: Dr. Young-Jun Park* — Seoul, Korea
- **George W. Woodruff School of Mechanical Engineering** — *Aug 2019 - Aug 2021*
  *Research Assistant | PI: Dr. Ye Zhao* — Atlanta, GA
- **Korean Institute of Machinery and Materials** — *Jun 2018 - Aug 2018*
  *Research Intern* — Daejeon, Korea

## PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, PAT=PATENT, S=IN SUBMISSION, P=PREPARATION, T=THESIS

**[C.1]** **D.H. Han**, X. Xu, Y. Chen, Y. Zhou, X. Zhang, D. Bruder, X. Huang. “Towards Whole-Body Proprioceptive Morphing: A Soft Gripper for Cross-Scale Grasping”. 2026 IEEE International Conference on Soft Robotics (RoboSoft).

**[C.2]** R. Zuo, M. Mehta, **D.H. Han**, D. Bruder. "Embedded Valves for Distributed Control of Soft Pneumatic Actuators". 2024 IEEE International Conference on Intelligent Robots and Systems (IROS)

**[C.3]** **D.H. Han**, S.J. Byeon, K.D. Kim, G.H. Han, M.H. Cha, Y.J. Park. "Development of Path Tracking Control Algorithm for Tractor Autonomous Driving". 2021 Korean Society for Agricultural Machinery Conference

**[Pat.1]** **Blowers With Variable Nozzles**. US 11668311 B2. Issued June 6, 2023.

**[S.1]** **D.H. Han**, M. Mehta, R. Zuo, Z. Wanger, and D. Bruder. "An Enhanced Proprioceptive Method for Soft Robots Integrating Bend Sensor and IMU"

**[S.2]** R. Zuo, **D.H. Han**, R Li, S. Jamal, D. Bruder. “UMArm: Untethered, Modular, Wearable, Soft Pneumatic Arm with 12 DoF”

**[S.3]** **D.H. Han**, R. Zuo, C. Kim, S. Jamal, D. Bruder. “BendSense: A Modular Off-the-Shelf Bend Sensor Framework for Proprioception in Soft Robots”

## HONORS AND AWARDS

- **VIP Innovation Competition, 1st Place in Hardware, Devices & Robotics Track** — *Apr 2021*
  *Georgia Institute of Technology*
  ◦ Awarded to the most innovative and active research team in Georgia Tech
- **President’s Undergraduate Research Award** — *Oct 2020*
  *Georgia Institute of Technology*
  ◦ Research excellence scholarship as an undergraduate researcher at Georgia Tech
- **Georgia Korean American Grocers Association Scholarship Award** — *Dec 2016*
  *KAGRO*
  ◦ Awarded for academic excellence and leadership in community service

## CORE SKILLS

- **Robotics:** System-level robotic design and integration, hardware/firmware/software co-design, ROS2
- **Programming:** C, C++, Python, MATLAB
- **Design and Fabrication:** Embeded System, CAD, 3D printing, silicone casting, rapid prototyping